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FocuS-MN: Focusing on Underwater Signal Denoising via Sequential Memory Networks with Learnable Resampling

Shouao Gu, Zitong Li and Jun Tang *

School of Civil Engineering, Tianjin University, Tianjin 300350, China; 2023205140@tju.edu.cn (S.G.); zitongli2003@tju.edu.cn (Z.L.)
* Correspondence: tangjun@tju.edu.cn

Abstract

The coupling of non-stationary marine noise and complex ship-radiated signals makes high-fidelity signal recovery exceptionally difficult. Existing deep learning methods often prioritize objective metrics, such as the Scale-Invariant Signal-to-Noise Ratio (SI-SNR), but fail to maintain the integrity of narrow-band line spectral data. We propose FocuS-MN, an end-to-end framework that combines learnable resampling with Feedforward Sequential Memory Network (FSMN)-based temporal modeling for precise waveform reconstruction. The model is optimized using a two-stage training strategy to ensure stable magnitude estimation and waveform consistency. On the ShipsEar dataset, FocuS-MN shows strong generalization to unseen vessel types. At a −5 dB Signal-to-Noise Ratio (SNR), it achieves a Signal-to-Distortion Ratio (SDR) of 3.77 dB and a Segmental Signal-to-Noise Ratio (SSNR) of 3.83 dB. Power Spectral Density (PSD) analysis further confirms that FocuS-MN recovers fine-grained line spectral structures, proving its effectiveness in both noise suppression and signal fidelity.

Keywords: underwater signal denoising; ship-radiated noise; Feedforward Sequential Memory Network (FSMN); time–frequency resolution

1. Introduction

Underwater acoustic signals are fundamental to underwater communication, resource exploration, and the detection, localization, and identification of underwater targets—such as submarines, autonomous underwater vehicles, and naval mines—representing the only viable modality for long-range sensing. The reliability of underwater acoustic signals directly underpins maritime defense capabilities and strategic naval initiatives. However, the inherent complexity of ship-radiated noise and the presence of diverse interference sources in experimental settings pose significant challenges. In highly sophisticated marine environments, including natural noise from wind, waves, currents, and rain [1,2], underwater acoustic signals are susceptible to the simultaneous influence of multiple physical variables [3]. Such interferences not only compromise the accuracy of target detection and identification but also exacerbate the difficulty of localization tasks [4]. To address these issues, denoising measures are imperative to suppress background noise and accentuate the features of ship-radiated noise. The focus of this paper is to propose an efficient methodology that not only removes ambient noise effectively but also achieves high-fidelity recovery of the low-frequency line spectral features inherent in original target vessel signatures.

Conventional methods such as spectral filtering, adaptive filtering, wavelet transform, and modal decomposition achieve denoising by modeling salient features and noise

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interference in the time–frequency domain. These approaches provide the foundational technical framework for the denoising and enhancement of underwater acoustic signals. Methods such as Wiener filtering [5] and spectral subtraction [6] are based on the statistical properties of noise in the frequency domain. These techniques achieve efficient denoising by modeling noise under the assumption of stationarity. However, such methods face limitations when actual underwater acoustic signals encounter multiple coupled noise types or non-stationary interference. Other approaches, including Empirical Mode Decomposition (EMD) [7–9], Ensemble Empirical Mode Decomposition (EEMD) [7,10], and Variational Mode Decomposition (VMD) [11–13], are used to perform modal decomposition on underwater acoustic signals. These techniques decompose the target signal into a sum of multiple Intrinsic Mode Functions (IMFs) across different frequency bands. Denoising is subsequently realized by selecting effective modes through thresholding or by discarding mode functions that contain noise. For instance, EMD cannot specify the number of modes, which often leads to mode mixing and the subsequent loss of effective features in underwater acoustic signals. Although EEMD effectively addresses the mode mixing problem, it introduces Gaussian white noise during the decomposition process. If this noise is not fully canceled out, additional denoising methods may be required for further processing. While VMD utilizes a variational approach to solve for each mode—offering a superior decomposition performance and robustness to some extent—its denoising effectiveness remains heavily dependent on parameters such as the predefined number of modes determined by empirical or a priori methods. The Adaptive Linear Enhancer (ALE) adopts the minimum mean difference adaptive algorithm, which is specially designed to extract the periodic line spectral signals from the broadband noise background in order to remove noise and enhance narrow-band underwater acoustic characteristics. However, ALE is not only weak against broadband noise, but also weak against the nonlinear modulation of high-frequency shocks [14]. In general, traditional underwater acoustic signal noise reduction techniques for nonlinear, non-stationary noise are limited and cannot handle underwater acoustic signal components.

Recently, speech denoising technologies based on Deep Neural Networks (DNNs) have demonstrated performance advantages over conventional methods in complex acoustic scenarios through end-to-end learning mechanisms. These technologies are capable of not only isolating target sound sources from environmental reverberation but also effectively suppressing short-term transient noise and wideband impulsive noise. Although underwater acoustic signals and speech signals exhibit significant differences in frequency distribution and propagation attenuation characteristics, the core technical framework of speech denoising still provides an important reference for the processing of underwater acoustic signals.

Early research on DNN-based denoising aimed to integrate prior knowledge from conventional signal processing with deep learning architectures. For instance, RNNoise [15] ingeniously incorporates the characteristics of human auditory sensitivity across different frequency bands into a lightweight Gated Recurrent Unit (GRU) network, utilizing traditional gain filtering to achieve high real-time processing efficiency. NSNet [16] further validated the robustness of learning based on signal masking properties within communication systems.

With the continuous advancement of computational power, studies such as [17,18] have shifted focus toward purely data-driven high-order temporal modeling methods. Specifically, Ref. [17] achieves the joint tracking of magnitude and phase evolution through complex-valued convolutional recurrent networks, effectively addressing the phase distortion issue in the reconstruction of non-stationary underwater acoustic signals. In [19], the simultaneous capture of global long-range dependencies and local transient features

is enhanced by deeply integrating full-band and sub-band information. Furthermore, Refs. [18,20] leverage the global perception capabilities of Transformers and adversarial learning mechanisms, respectively, to increase the representation depth of long-sequence waveforms to the limit.

At the same time, however, the denoising of underwater acoustic signals faces more severe cross-domain challenges. First, the disparity in physical representation. The core signatures of ship-radiated signals are concentrated in unique, low-frequency, narrow-band line spectral data. These features demand exceptional frequency resolution, as their spectral distribution deviates significantly from that of human speech. Consequently, generic speech models often fail to capture these fine-grained structures, leading to the severe distortion or loss of critical signatures. On the other hand, while speech enhancement focuses on suppressing background noise, such as street clamor or indoor reverberation, underwater noise reduction aims to eliminate ambient sea noise that masks the signals of target vessels. This requirement for high-precision restoration under extreme low-SNR conditions necessitates a departure from traditional speech technology toward a specialized paradigm centered on line spectral fidelity and structural reconstruction.

To address the aforementioned issues, the primary contributions of this paper are as follows:

- A multi-feature input scheme is proposed to address the complex coherence of underwater signals, utilizing a dual-stream parallel encoder to process magnitude and phase spectra simultaneously. This architecture enhances the model's perception of non-stationary time–frequency characteristics while preserving essential phase information, providing a robust physical basis for high-precision waveform reconstruction and improving signal restoration performance in low-SNR environments.
- A multi-layer residual Feedforward Sequential Memory Network (FSMN) architecture is introduced to capture long-range temporal dependencies through its unique memory block mechanism, addressing the limitations of conventional denoising in complex noise modeling. By iteratively separating background interference across layers, this architecture effectively compensates for the spectral fragmentation typically caused by time–frequency convolutional methods when processing quasi-periodic signals. This ensures the structural stability and continuity of critical line spectral in ship-radiated noise.
- A bidirectional frequency–time resampling mechanism is designed within the core layers of the model. By performing downsampling before the FSMN to compress redundant information and highlight global noise patterns, followed by precise upsampling at the output to restore the original resolution, the framework achieves an optimal balance between noise modeling and signal detail preservation. This approach effectively reduces computational overhead while ensuring high-fidelity reconstruction of the target audio.
- Extensive experiments on the ShipsEar dataset demonstrate that FocuS-MN outperforms both conventional enhancement methods and representative deep learning architectures, particularly in challenging scenarios at -5 dB SNR. Furthermore, ablation studies validate the critical roles of the resampling mechanism and FSMN modules in improving objective metrics and maintaining line spectral integrity. These results provide strong evidence of the model's effectiveness in the high-fidelity reconstruction and robust denoising of underwater acoustic signals.

2. Related Work

2.1. DNN-Based Denoising

The classical theoretical foundation of audio denoising originates from the analysis of signal statistical properties. It is assumed that the noisy signal $y(t)$ is a linear superposition of the target signal $x(t)$ and the additive environmental noise $n(t)$ in the time domain as follows:

$$y(t) = x(t) + n(t) \tag{1}$$

where t denotes the discrete time index.

Traditional denoising methods, such as spectral subtraction, typically rely on prior assumptions regarding noise stationarity. However, in complex maritime environments, noise often exhibits intense non-stationarity. The introduction of Deep Neural Networks (DNNs) has shifted this paradigm; its fundamental objective is to construct an end-to-end nonlinear mapping function from the noisy signal $y(t)$ to the target signal $x(t)$.

In the evolution of DNN-based denoising, several milestone architectures established in the field of speech enhancement have gradually formed a solid paradigm for general audio denoising. RNNoise [15] achieves lightweight processing by combining traditional features with recurrent networks. SEGAN [21] introduces Generative Adversarial Networks (GANs) to implement the direct reconstruction of time domain waveforms. DC-CRN [17] utilizes complex-valued convolutional networks for the effective extraction of phase information, while FullSubNet [19] enhances the model's ability to capture both global envelopes and local details through a combined full-band and sub-band mechanism. Furthermore, MP-SENet [20] implements parallel modeling of magnitude and phase spectra via a multi-path structure, demonstrating exceptional fidelity in complex acoustic environments.

Inspired by the above research, the denoising of underwater acoustic signals has gradually shifted from general architectures to deep customization tailored to the statistical properties of the marine environment. In the early stages of architectural evolution, researchers established a foundation by migrating mature models from the speech domain, such as utilizing the symmetrical structure of U-Net [22] to implement multi-scale time–frequency mapping or employing Bidirectional Denoising Autoencoders (BDAEs) [23], to enhance the feature discriminability of weak target signals. Furthermore, robust algorithms with non-Gaussian noise processing capabilities have been designed specifically for marine impulsive interference [24]. As the demand for environmental adaptability has grown, noise-aware models [25] have emerged, endowing networks with the capability to dynamically perceive variations in underwater environments. Concurrently, the introduction of dual-path Transformers [26] has further strengthened the capture of long-range dependencies in complex acoustic fields through joint modeling of the temporal and spatial axes. Recently, the research focus has shifted toward refined feature decoupling and high-quality reconstruction. For instance, DBSA-Net [27] achieves the separation of broadband and narrow-band features through a dual-branch self-attention mechanism, significantly improving accuracy under low-SNR conditions. In the field of Generative Adversarial Networks (GANs), based on advancements in noise modeling [28], researchers have further explored denoising GANs [29] that combine magnitude and phase spectra. This approach effectively mitigates the spectral distortion defects inherent in traditional methods during signal reconstruction.

2.2. Temporal Modeling Methods and FSMN

In the architectural design of deep learning models for audio signal processing, temporal modeling techniques have, overall, undergone an evolution from recurrent mechanisms to global self-attention mechanisms.

The recurrent modeling path, represented by Recurrent Neural Networks (RNNs) and their variants such as LSTM [30] and GRU [31], establishes temporal dependency representations of time domain information through the sequential transfer of hidden states. This mechanism plays a vital role in classic denoising architectures like RNNoise [15] and DCCRN [17], effectively processing speech components with high temporal correlation. During processing, the recurrent structure compresses historical features into the current state via point-by-point iteration, thereby achieving dynamic tracking of continuous signals.

With the rise of the Transformer [32] architecture, models based on the self-attention mechanism have become another mainstream approach for temporal modeling. Unlike the point-by-point iteration of recurrent structures, self-attention captures global information across long sequences by calculating correlation weights between any two positions within the sequence. Currently, this mechanism has demonstrated a superior performance in speech tasks [33–36], becoming a core architecture for audio temporal modeling and audio representation learning. It exhibits robust feature representation capabilities when processing signals under complex background noise.

Feedforward Sequential Memory Networks (FSMNs) represent another evolutionary path in temporal modeling, combining the parallelism of convolutional structures with the memory characteristics of recurrent structures [37]. Its core innovation lies in the design of memory blocks. When first proposed, FSMN was primarily utilized for front-end speech representation extraction in Automatic Speech Recognition (ASR) tasks. Specifically, FSMN bypasses the constraints of traditional recurrent units by employing a non-recurrent structure to capture temporal dependencies, the mechanism of which is detailed as follows.

For the l-th hidden layer, assume the input sequence is denoted as $H^{(1)} = \{h_1, h_2, \ldots, h_T\}$. The fundamental concept of FSMN is to encode the contextual information surrounding the hidden layer at time step t by utilizing a set of learned, time-invariant scalar coefficients.

For an N-th order FSMN model, the process of obtaining contextual representations is as follows. Assume a time series $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$ of length T, where the input at each time step t is $\mathbf{x}_t$. The corresponding output of this sequence through the l-th hidden layer is denoted as $\mathbf{H}$, which similarly has a corresponding vector $\mathbf{h}_t$ at each time step t. The objective of FSMN is to encode $\mathbf{h}_t$ at position t within the hidden layer output using a set of $N + 1$ scalar coefficients, thereby forming a fixed-size representation.

$$\mathbf{h}_t^1 = \sum_{i=0}^{N} \alpha_i \cdot \mathbf{h}_{t-i}^1 \tag{2}$$

In this framework, α denotes the time-invariant coefficients, which are used to weight the previous N time steps and sum them with $\mathbf{h}_t$ to obtain the representation of the current time step, thereby completing the memory of the aforementioned features. This process is analogous to the structure of a Finite Impulse Response (FIR) filter, where a set of time-invariant coefficients a denotes the system parameters of the filter, performing a 1D convolution with the input sequence to achieve temporal modeling. By relying on this structure, the model can capture not only historical information but also incorporate future information to form a bidirectional FSMN.

$$\hat{\mathbf{h}}_t = \sum_{i=0}^{N} \alpha_i \cdot \mathbf{h}_{t-i}^l + \sum_{j=1}^{M} \beta_j \cdot \mathbf{h}_{t+j}^l \tag{3}$$

β denotes the time-invariant coefficients for future information, with its weighted summation performed in the same manner as the unidirectional FSMN. Finally, the temporal representation that accounts for global context is calculated by summing the two individual representations.

2.3. Signal Resampling and Multi-Scale Architectures

In the task of underwater acoustic signal enhancement, the core challenge lies in balancing the computational burden of long-range dependency modeling with the fidelity of time–frequency details. Recently, constructing multi-scale architectures by introducing explicit resampling mechanisms within models has become a mainstream trend for processing complex temporal signals.

Early waveform-based encoder–decoder models, such as Wave-U-Net [38], employed explicit temporal resolution changes to build hierarchical representations. In Wave-U-Net, downsampling is implemented via simple decimation (subsampling), while upsampling uses linear interpolation followed by convolutional refinement. This approach avoids learnable transposed convolutions, thereby reducing checkerboard artifacts in the reconstructed waveform.

Subsequent models, including Demucs [39], adopted strided convolutions for downsampling and transposed convolutions for upsampling, forming a fully learnable hierarchical resampling structure. However, transposed convolutions are known to introduce severe spectral artifacts in waveform synthesis [40], which can be particularly detrimental for underwater target signals that require high-fidelity reconstruction.

In these architectures, temporal resolution reduction primarily serves to expand the receptive field and enable multi-scale feature extraction, but the resampling schedule remains fixed and manually designed. This limitation motivates the later shift toward learned compression in neural codecs and implicit temporal resolution control in modern architectures.

Neural audio codecs, such as SoundStream [41] and Encodec [42], are designed to ensure high-fidelity signal reconstruction. To achieve more efficient compression in the latent space, these models incorporate learnable resampling operations within their encoder–decoder structures, allowing adaptive manipulation of temporal resolution while preserving essential signal information. Unlike the fixed downsampling in traditional encoder–decoder architectures, such mechanisms enable more flexible and efficient representation learning.

More recent architectures move beyond explicit resampling and instead adopt implicit temporal resolution manipulation strategies. Efficient Conformer [43], Squeezeformer [44], Audio mamba [45] and Zipformer [35] reduce computational costs through various scale-compression mechanisms. In particular, Zipformer introduces a learnable compression layer that dynamically merges similar temporal representations without relying on fixed rules like pooling or strided downsampling. Audio-Mamba, on the other hand, combines the input features obtained through patch embeddings and performs hierarchical downsampling via patch merging, thereby constructing a multi-level representation of the audio.

In this paradigm, temporal resolution is no longer treated as a fixed structural ladder nor solely as a compression bottleneck, but as an implicit and dynamically allocated modeling resource. Such a perspective reflects a conceptual shift: resolution becomes part of representation control rather than a predefined sampling schedule.

3. Proposed FocuS-MN Architecture

The overall architecture of the model is illustrated in Figure 1, comprising front-end feature extraction, a symmetric encoder–decoder structure with integrated downsampling/upsampling layers, and FSMN blocks for temporal modeling, ultimately terminating in a magnitude spectrum decoder. This structure is designed to map noisy signals to target signals, thereby achieving signal reconstruction. Notably, the model design incorporates two types of stacking configurations: one involves the independent stacking of FSMN blocks, while the other involves the joint stacking of up/downsampling operations with

FSMN blocks. This dual-stacking strategy enhances the model's capability for feature modeling across various time scales and frequency bands.

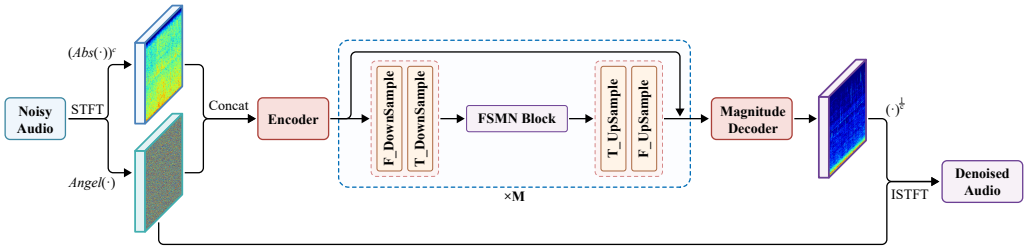


Figure 1. The architecture of FocuS-MN. The framework employs an end-to-end processing pipeline starting with STFT to extract magnitude and phase information. The concatenated features pass through an encoder, followed by M stages of a symmetrical resampling structure consisting of frequency/temporal downsampling, FSMN blocks for latent temporal modeling, and corresponding upsampling modules. Finally, the magnitude decoder and ISTFT are utilized to reconstruct the denoised audio.

3.1. Front-End

At the front-end signal processing stage, the workflow of this study is as follows:

The front-end processing workflow of this study aims to transform one-dimensional time domain signals into two-dimensional time–frequency features. First, the signals are decomposed into the frequency domain using the Short-Time Fourier Transform (STFT).

$$X(m,k) = \sum_{n=0}^{N-1} x(n+mH)w(n)e^{-j\frac{2\pi}{N}kn} \tag{4}$$

In this process, $X(m,k)$ denotes the extracted complex spectrum. Here, m and k represent the time frame index and frequency bin index, respectively. n is the local time index within a frame of length N. The term $w(n)$ denotes the window function; in this study, a Hanning window is employed to reduce spectral leakage. The parameter H refers to the hop size, which determines the step interval between successive STFT frames. The STFT of the input signal yields a complex spectrum $X(m,k)$. For notational conciseness, we omit the indices (m,k) and denote the spectrum as X where the context is clear. Subsequently, the magnitude spectrum Y_m and the wrapped phase spectrum Y_p are extracted as follows:

$$Y_m(m,k) = |X(m,k)| = \sqrt{\mathrm{Re}^2(X) + \mathrm{Im}^2(X)} \tag{5}$$

$$Y_p(m,k) = \angle X(m,k) = \arctan 2\big(\mathrm{Im}(X), \mathrm{Re}(X)\big) \tag{6}$$

In this context, $\mathrm{Re}(\cdot)$ and $\mathrm{Im}(\cdot)$ represent the real and imaginary components, respectively. Y_m denotes the magnitude spectrum, characterizing the distribution intensity of energy across the time–frequency plane, while Y_p denotes the phase spectrum, which encapsulates the temporal coherence of different frequency components. Furthermore, considering the significant variations in the dynamic range of underwater acoustic signals, we apply power law nonlinear compression to the magnitude spectrum Y_m before it is fed into the encoder:

$$Y_{cm} = Y_m^p \tag{7}$$

where p denotes the compression coefficient. The compressed features exhibit a numerical distribution that is more closely normalized, which facilitates the accelerated convergence of the deep neural network and alleviates the issue of gradient distortion. After completing the feature construction of the magnitude and phase spectra, this study concatenates the

two along the channel dimension to form a joint time–frequency representation as the model input.

3.2. Encoder

The structure of the encoder is illustrated in Figure 2. Inspired by MP-SENet [20], our encoder is designed to process the concatenated magnitude and phase spectra features, capturing long-range time–frequency dependencies through convolutional blocks and Dilated DenseNet modules. Specifically, the encoder consists of the following components:

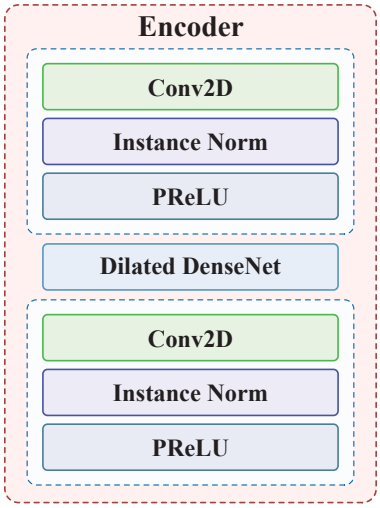


Figure 2. Components of the Dilated DenseNet-based encoder. The encoder employs a hierarchical architecture consisting of Conv2D and dilated convolutional layers designed for high-resolution feature extraction. Every convolutional operation is coupled with Instance Normalization and PReLU activation functions.

Initial Convolutional Block

This block employs a 2D convolutional layer with a 3×3 kernel and a stride of 1×1 to map the two-channel input features into 32 channels, followed by Instance Normalization and PReLU activation.

Dilated DenseNet Module

This module is composed of DenseDilatedLayers, where each layer utilizes dilated convolutions in both temporal and frequency directions. The dilation rates are sequentially set to $(1, 2, 4, 8)$, and each layer is followed by Instance Normalization and PReLU activation. This module is capable of modeling long-range temporal and spectral dependencies.

Output Convolutional Block

A convolutional layer with a 3×3 kernel and a stride of 1×1 is used, along with Instance Normalization and PReLU activation, to map the feature channels to 32 for subsequent processing. Unlike the original MP-SENet, this encoder does not compress the frequency dimension (the frequency scaling factor is set to 1). This design preserves high-frequency details, which facilitates the capture of transient impulses and sharp spectral variations in underwater noise signals, thereby enhancing the model's ability to model high-frequency transient features.

3.3. Resampling

In the architecture of the noise reduction model, to flexibly adjust the time–frequency resolution and reduce computational complexity, this model introduces a learnable down-sampling module and an upsampling module, acting on the frequency and time dimensions,

respectively. Through the downsampling module, the original features are compressed from (B, C, T, F) to $(B, C, T/2, F/2)$, and subsequently restored to (B, C, T, F) via the upsampling process. The primary logic is detailed in Algorithms 1 and 2:

Algorithm 1 Learnable Adjacent-Frame Downsampling

Input: Feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times D\times *}$, learnable weights $\mathbf{W} \in \mathbb{R}^{C\times 2}$
Output: Downsampled feature map $\mathbf{Y} \in \mathbb{R}^{B\times C\times \frac{D}{2}\times *}$
Normalize weights using softmax:
  $\boldsymbol{\alpha}_c = \mathrm{Softmax}(\mathbf{W}_c),\ c = 1, \ldots, C$
(Optional) Pad $\mathbf{X}$ along dimension D to ensure even length
for each channel $c = 1, \ldots, C$ **do**
  for each index $k = 1, 2, \ldots, \lfloor D/2 \rfloor$ **do**
    $\mathbf{Y}_{:,c,k,:} \leftarrow \frac{1}{2}(\alpha_{c,0}\mathbf{X}_{:,c,2k-1,:} + \alpha_{c,1}\mathbf{X}_{:,c,2k,:})$
  end for
end for

Algorithm 2 Adjacent-Frame Upsampling

Input: Feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times D\times *}$
Output: Upsampled feature map $\mathbf{Y} \in \mathbb{R}^{B\times C\times 2D\times *}$
for each channel $c = 1, \ldots, C$ **do**
  for each index $k = 1, 2, \ldots, D$ **do**
    $\mathbf{Y}_{:,c,2k-1,:} \leftarrow \mathbf{X}_{:,c,k,:}$
    $\mathbf{Y}_{:,c,2k,:} \leftarrow \mathbf{X}_{:,c,k,:}$
  end for
end for

Downsampling Module

This module performs compression across the time–frequency domain. Each channel is assigned learnable weighting parameters, and the feature downsampling is achieved by calculating a weighted average of adjacent temporal frames and frequency bins.

Upsampling Module

This module restores the features to their original resolution by repeating each temporal frame and frequency bin. The introduction of the learnable downsampling module not only compresses the time–frequency resolution and alleviates the computational burden but also enables the modeling of temporal and spectral features at a lower frame rate, thereby enhancing the model's ability to capture long-range time–frequency patterns. The upsampling module provides a simple and efficient means of restoration, allowing the encoder–decoder structure to improve overall modeling capacity while maintaining information integrity.

3.4. FSMN Block

The structures of the FSMN block and the FSMN basic block are illustrated in Figure 3 and Figure 4, respectively. After the spectrogram is fed into the model, it first passes through two layers of affine transformations. These layers compress the high-dimensional spectral features into a lower-dimensional space that facilitates the extraction of contextual representations while reducing redundant computation.

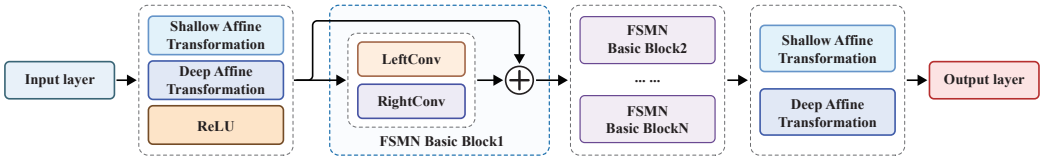


Figure 3. Detailed architecture of the FSMN block. The modeling process includes affine transformations and ReLU activations preceding a sequence of N FSMN basic blocks. Each block utilizes a residual-like structure with "LeftConv" and "RightConv" components to fuse past and future temporal context. This hierarchical arrangement enables the model to perform multi-scale feature extraction and fine-grained noise suppression in the latent space.

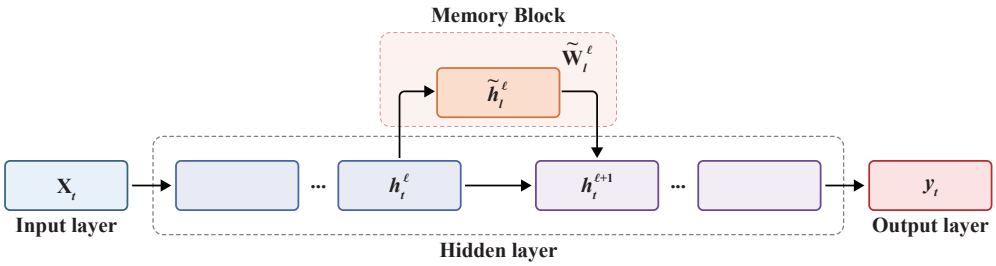


Figure 4. Internal structure of the FSMN basic block. This diagram illustrates the memory mechanism utilized within the hidden layers of the network. The current hidden state h_t^{ℓ} is coupled with a memory block that accumulates past and future context using learned coefficients $\tilde{\mathbf{W}}_t^{\ell}$. This architecture allows the model to capture long-range temporal dependencies more efficiently than standard RNNs, facilitating precise noise pattern recognition within the latent representation space.

FSMN Basic Block Architecture

The first FSMN basic block receives the dimensionality-reduced spectral information and refines the spectral focus through a linear layer. Correspondingly, the final layer of each basic block is also a linear layer, which restores the features to their original input shape to complete the output reconstruction. Between these two linear layers, the FSMN basic block introduces the learnable memory module mentioned in Section 2.2.

Contextual Modeling

By utilizing 1D convolutions across independent channels, the module captures and stores fixed-length historical and future information within different frequency bands. This forms bidirectional information coverage for the contextual modeling of acoustic samples. Assuming the features after affine transformation, X, are input into the FSMN basic block, left-side padding is applied along the time steps to maintain a constant temporal length; the result after padding is denoted by X_{lpad}.

$$\mathbf{H}_{left} = \text{Conv2D}\left(\mathbf{X}_{lpad}, \mathbf{W}_{left}\right), \mathbf{W}_{left} \in \mathbb{R}^{F \times 1 \times L_{order} \times x_1} \tag{8}$$

In this formulation, $\mathbf{H}_{left}$ denotes the representation resulting from modeling the preceding context using L_{order} historical time steps, and $\mathbf{W}_{left}$ denotes the convolution kernel for historical information.

Similarly, for the modeling of future information, the computation is identical to that of the left-side convolution, with the exception that the feature $\mathbf{X} \in \mathbb{R}^{T \times C}$ is first padded on the right side to ensure the convolution kernel covers the required future time steps:

$$\mathbf{H}_{right} = \mathbf{X}_{rpad} \times \mathbf{W}_{right} \tag{9}$$

where $\mathbf{H}_{right}$ denotes the representation of future context modeling and $\mathbf{W}_{right}$ denotes the corresponding future information convolution kernel. By combining these two components,

the FSMN basic block achieves a symmetric and comprehensive receptive field across the temporal axis. This dual-path architecture ensures that transient acoustic events are captured alongside long-term steady-state signals.

$$\mathbf{H}_{right} = \text{Conv2D}\left(\mathbf{X}_{rpad}, \mathbf{W}_{right}\right), \quad \mathbf{W}_{right} \in \mathbb{R}^{F\times 1\times L_{order}\times x_1} \tag{10}$$

Concurrently, a residual path is designed within the basic block to mitigate the network degradation problem. Specifically, the output of each layer is the summation of the input and the convolution results.

$$\mathbf{H}_{\text{out}} = \mathbf{X} + \mathbf{H}_{left} + \mathbf{H}_{right} \tag{11}$$

In essence, the FSMN basic block establishes a noise processing paradigm characterized by local filtering and global compensation. The 1D convolutions rapidly capture the locations of noise occurrences along the temporal steps, while the memory modules dynamically recover the abstract representations of narrow-band underwater acoustic features masked by noise. This dual-action approach enables the model to simultaneously address the continuous interference of broadband noise and the sudden disruption of transient noise. Furthermore, the model incorporates both FSMN stacks with resampling and stacks of basic blocks to construct the deep architecture of the FSMN model.

3.5. Magnitude Decoder

To achieve effective suppression of natural underwater noise, this study designs a lightweight magnitude mask decoder (or magnitude masking decoder) to estimate spectral magnitude masks from high-level time–frequency features. The decoder takes upsampled FSMN features as the input to adaptively suppress background noise while preserving target signal structural stability.

The structure of the mask decoder is illustrated in Figure 5. First, the decoder introduces a densely connected structure (Dense Block) to enhance the input features, strengthening local time–frequency correlations through feature reuse and improving the gradient propagation capability of the network. This design helps preserve the continuous spectral variation characteristics of natural underwater noise, preventing excessive smoothing or structural degradation.

Subsequently, a convolutional mask estimation module is employed to map the features. This module consists of transposed convolution, Instance Normalization (IN), and Parametric ReLU (PReLU) activation functions, which serve to refine the spectral structure along the frequency dimension and enhance nonlinear modeling capabilities. Specifically, Instance Normalization effectively mitigates the impact of magnitude distribution variances caused by different sea states and acquisition conditions, thereby improving the model's robustness across diverse environmental settings. Finally, a point-wise convolution maps the features into a single-channel magnitude mask representation. At the mask output stage, this study adopts a frequency-adaptive learnable Sigmoid function to constrain the magnitude mask, ensuring that mask values remain within a reasonable range. Unlike a fixed Sigmoid activation, this learnable mechanism allows the model to adaptively adjust the dynamic range of the mask for different frequency bands. This enables a more accurate characterization of the non-uniform distribution properties of natural underwater noise across the spectrum.The mask is defined as

$$M = \beta \cdot \sigma(S \odot X) \tag{12}$$

where X denotes the unconstrained features from the preceding decoder layer, S denotes a set of learnable scaling parameters, σ represents the Sigmoid function (i.e., $\sigma(x) = \frac{1}{1+e^{-x}}$), and β denotes the overall magnitude scaling coefficient for the mask. Finally, the magnitude mask output by the decoder is multiplied by the magnitude of the noisy spectrogram to obtain the enhanced magnitude spectrum representation, which is used for subsequent time domain signal reconstruction. This magnitude mask decoder achieves effective suppression of natural underwater noise while ensuring waveform continuity and physical plausibility.

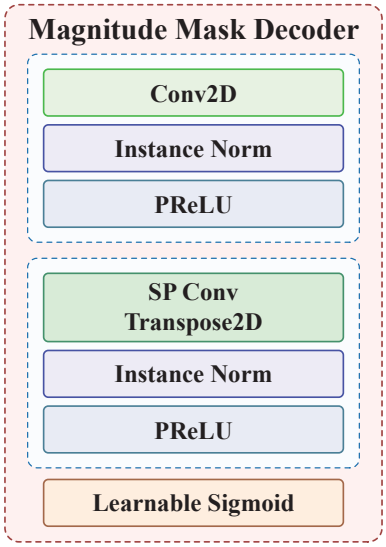


Figure 5. Components of the magnitude mask decoder. The decoder utilizes a multi-layer structure comprising Conv2D and SP Conv Transpose2D layers for feature restoration. Each convolutional stage is followed by Instance Normalization and PReLU activations. A learnable Sigmoid function is applied at the final stage to generate the magnitude mask, ensuring precise spectral reconstruction of the target signal.

4. Experiment

4.1. Dataset

The ShipsEar dataset [46] consists of high-fidelity underwater acoustic recordings of various ship types and pure background noise samples collected by hydrophones deployed in shallow water. This dataset is widely adopted in the field of underwater acoustic signal processing, providing a reliable data foundation for the modeling and analysis of ship-radiated noise.

In real marine environments, obtaining perfectly clean underwater target acoustic signals is challenging due to propagation loss, multi-path effects, and complex background noise. However, when hydrophones are deployed in close proximity to the target vessel, the captured signals typically exhibit a high Signal-to-Noise Ratio (SNR), where the influence of background noise is negligible to a certain extent, thus ensuring the high-fidelity characteristics of the ship's acoustic signal. The ship samples provided in the ShipsEar dataset were obtained under such close-range acquisition conditions, offering high data quality.

In this study, we selected acoustic samples of all ship categories from the ShipsEar dataset and chose typical marine environmental noises, such as wind, current, and rain noise from the background data, as interference signals. To unify the data format and reduce computational complexity, all ship signals and selected noise signals were downsampled to 16 kHz and cropped into audio segments with a duration of 3 s.

In the experimental setup, we do not consider the residual background noise components inherent in the original ship recordings, treating them as approximately "clean" target

signals. Subsequently, the selected background noise is superimposed onto the ship signals at an SNR of −5 dB to construct representative low-SNR underwater acoustic samples.

Regarding the dataset partition, all noisy samples from four ship categories—dredgers, tugs, trawlers, and pilot boats—were designated as the test set to evaluate the model's generalization performance on unseen ship categories. The noisy samples from the remaining seven ship categories constitute the development set, which was further randomly partitioned into a training set and a validation set. Ultimately, the ratio of samples in the training, validation, and test sets is approximately 7:2:1.

4.2. Experiment Steps

4.2.1. Experimental Setup

The experiments were implemented using the PyTorch (Version 2.4.1) framework and trained on an NVIDIA GeForce RTX 3090 (24 GB VRAM) platform. The primary configurations are as follows: the Adam optimizer was employed with a weight decay coefficient of 0.01; and the initial learning rate was set to 5×10^{-5} and dynamically adjusted via a cosine annealing scheduler. To prevent overfitting, an early stopping mechanism was implemented with a patience of 20, terminating the training if the validation loss failed to decrease for 20 consecutive epochs. The batch size was fixed at 16 to ensure controllable VRAM consumption, with a maximum training limit of 200 epochs.

4.2.2. Loss Function and Training Strategy

To ensure that the model achieves stable and precise reconstruction, this study adopts a two-stage training strategy. This approach is designed to alleviate optimization conflicts between different representations and enhance the overall stability of convergence.

Stage I: Magnitude Pre-training (Magnitude Reconstruction Stage)

In the first stage, the optimization focuses exclusively on magnitude modeling. This stage involves comprehensive modeling of magnitude information by minimizing the Mean Squared Error (MSE) loss function between the predicted magnitude spectrum and the clean magnitude spectrum:

$$L_{mag} = \frac{1}{T \times F} \sum_{t=1}^{T} \sum_{f=1}^{F} \left(|Y(t,f)| - |\hat{Y}(t,f)| \right)^2 \tag{13}$$

where $|Y(t,f)|$ and $|\hat{Y}(t,f)|$ denote the magnitude values of the clean and enhanced spectrograms at time t and frequency f, respectively. This pre-training phase allows the encoder and the magnitude mask decoder to prioritize capturing the structural characteristics of the ship-radiated noise without the interference of phase optimization.

This phase aims to enable the model to prioritize learning stable energy distributions and time–frequency structures, thereby providing a reliable magnitude prior for subsequent phase modeling.

Stage II: Joint Magnitude and Waveform Modeling

In the second stage, the SI-SNR (Scale-Invariant Signal-to-Noise Ratio) loss is employed to directly constrain the temporal similarity between the reconstructed waveform and the reference waveform. This enables the model to further optimize perceptual quality and signal integrity, achieving an overall enhancement in consistency from frequency domain modeling to time domain reconstruction.

The SI-SNR loss is defined as follows:

$$\text{SI-SNR loss} = -10 log_{10} \frac{||x_{target}||^2}{||e_{noise}||^2} \tag{14}$$

where the target signal $\mathbf{x}_{target}$ and the residual noise $\mathbf{e}_{noise}$ are calculated as

$$x_{target} := \frac{< \hat{x}, x > x}{||x||^2}, e_{noise} := \hat{x} - x_{target} \tag{15}$$

4.2.3. Evaluation Metrics

Regarding evaluation metrics, we select Signal-to-Distortion Ratio (SDR), Signal-to-Distortion Ratio improvement (SDRi), and Segmental Signal-to-Noise Ratio (SSNR) as the quantitative evaluation indicators.

Signal-to-Distortion Ratio (SDR)

SDR is a comprehensive metric used to evaluate the overall quality of the enhanced signal by considering the ratio of the target signal energy to the total error energy (including interference, artifacts, and residual noise). A higher SDR value indicates a better separation performance and lower distortion.

$$SDR = 10 \log_{10} \frac{\sum_{n=1}^{L} x^2(n)}{\sum_{n=1}^{L} [x(n) - \hat{x}(n)]^2} \tag{16}$$

SDR Improvement (SDRi)

SDRi represents the net gain in SDR after processing, defined as the difference between the SDR of the enhanced signal and the SDR of the original noisy signal as follows:

$$\text{SDRi} = \text{SDR}_{enhanced} - \text{SDR}_{noisy} \tag{17}$$

This metric effectively isolates the performance of the model from the initial quality of the input samples, providing a direct measure of the noise suppression efficiency.

Segmental Signal-to-Noise Ratio (SSNR)

Unlike the global SNR, SSNR calculates the Signal-to-Noise Ratio within short time frames (typically 15–30 ms) and averages them. This approach is more sensitive to the local variations in ship-radiated noise and avoids the bias caused by high-energy segments dominating the global ratio as follows:

$$\text{SSNR} = \frac{10}{M} \sum_{m=0}^{M-1} \log_{10} \frac{\sum_{n=n_m}^{n_m+N-1} s^2(n)}{\sum_{n=n_m}^{n_m+N-1} (s(n) - \hat{s}(n))^2} \tag{18}$$

where M denotes the total number of frames and N denotes the frame length.

Additionally, Power Spectral Density (PSD) analysis is introduced to measure the discrepancy in frequency distribution between the reconstructed and original signals. This enables an assessment of whether the frequency components of the target ship signal are damaged during the noise reduction process. We contend that PSD analysis facilitates the physical-level verification of whether the model effectively preserves the characteristic frequency components of ship-radiated noise while suppressing broadband impulsive noise.

By examining the PSD, it becomes possible to observe the preservation of line spectral and harmonic structures—the "acoustic fingerprints" of vessels—which are often buried under the continuous spectrum of natural marine noise. Maintaining the integrity of these components is crucial for downstream tasks such as passive sonar detection and target classification.

5. Results

5.1. Ablation Study Results

To further investigate the temporal modeling capability of the FSMN memory blocks and their synergistic effect with the resampling architecture (Downsampling–FSMN–Upsampling), this study designed a series of systematic ablation experiments. The objective was to quantitatively analyze the impact of temporal modeling depth across different dimensions on denoising performance. In the experiments, we configured the FocuS-MN model with varying numbers of layers, M and N, where M represents the number of stacked FSMN blocks containing resampling modules and N denotes the number of FSMN blocks deployed within the latent representation space after resampling.

We utilized the configuration $\{M = 4, N = 4\}$ as the baseline model. By cross-adjusting the values of M and N, we constructed six representative configurations: $\{M = 4, N = 4\}$, $\{M = 4, N = 6\}$, $\{M = 4, N = 8\}$, $\{M = 6, N = 6\}$, $\{M = 6, N = 4\}$, and $\{M = 8, N = 4\}$.

In terms of the evaluation framework, the experiments adhered to the standardized criteria used throughout this paper, employing the Signal-to-Distortion Ratio (SDR) and Segmental Signal-to-Noise Ratio (SSNR) as key metrics to measure denoising quality. These metrics aim to comprehensively evaluate the model's ability to maintain signal fidelity and suppress background noise under different configurations. The experimental results for SDR and SSNR under various configurations are presented in Table 1.

Table 1. Ablation study on the impact of FSMN block stacking layers (SDR/SSNR, dB).

Latent Space Layers (N)	**FSMN Layers with Resampling (M)**		
	M = 4	**M = 6**	**M = 8**
N = 4	3.56/3.63	3.73/3.79	3.57/3.64
N = 6	3.61/3.68	**3.77/3.83**	–
N = 8	3.63/3.68	–	–

Note: The symbol "–" indicates untested configurations. This study selects representative parameter points ($N, M \in \{4, 6, 8\}$) to characterize the evolutionary trend of model performance, which sufficiently reveals the nonlinear relationship between structural depth and enhancement efficacy.

As indicated in Table 1, the model performance exhibits a clear trend of initially increasing and then decreasing as the number of layers grows. The model reaches its performance peak when both M and N are set to six, achieving an SDR of 3.77 dB and an SSNR of 3.82 dB. Furthermore, compared to the $\{M = 4, N = 6\}$ and $\{M = 6, N = 4\}$ groups, the performance gain is most significant when the model is expanded from $\{M = 4, N = 4\}$ to $\{M = 6, N = 6\}$, with the SDR increasing from 3.56 dB to 3.77 dB. This suggests that both M and N achieve effective stacking and form a deep synergy, efficiently modeling the noisy signals within the dataset. Conversely, the "unilateral deepening" experiments, $\{M = 8, N = 4\}$ and $\{M = 4, N = 8\}$, experienced a degree of degradation and failed to deliver the expected stacking gain, showing marginal improvements over the $\{M = 4, N = 4\}$ baseline.

Based on these results, we conclude that the $\{M = 6, N = 6\}$ configuration, through its symmetrical architectural design, achieves the optimal balance between intra-resolution perception and overall temporal modeling. Therefore, this study adopts this configuration as the final model architecture for all subsequent experiments.

5.2. Comparative Results

To verify the effectiveness of the proposed method in denoising tasks, this study selects two classic signal decomposition-based methods and three representative deep learning models as comparative baselines.

Traditional Signal Processing Methods

We employ Empirical Mode Decomposition (EMD) and Variational Mode Decomposition (VMD). These methods are widely utilized in the fields of non-stationary signal analysis and underwater acoustic signal processing, representing the traditional benchmark for handling complex acoustic signals.

Deep Learning Methods

Furthermore, three representative deep learning architectures are selected for comparison:

DCCRN (Deep Complex Convolutional Recurrent Network): A complex-valued network designed to process both magnitude and phase information simultaneously.

SEGAN (Speech Enhancement Generative Adversarial Network): An end-to-end generative model that operates directly in the time domain.

FullSubNet: A model that utilizes both full-band and sub-band information to capture global and local spectral patterns.

5.2.1. Comparison with Traditional Methods

Empirical Mode Decomposition (EMD) and Variational Mode Decomposition (VMD) are both adaptive modal decomposition methods. Their core principle involves decomposing complex non-stationary signals into several Intrinsic Mode Functions (IMFs) with narrow-band characteristics. Given that underwater acoustic signals typically exhibit significant non-stationarity and multi-component superposition, modal decomposition methods have demonstrated good applicability to acoustic signal analysis and denoising tasks. Despite differences in their specific decomposition mechanisms, EMD and VMD share a consistent modeling philosophy—suppressing noise effectively at the component level. Therefore, they are selected as the representative traditional signal processing benchmarks in this study.

In our experiments, both EMD and VMD follow a standard "Decomposition–Selection–Reconstruction" denoising pipeline. For EMD, the first m high-frequency IMF components are discarded during the reconstruction phase, and the remaining IMFs are aggregated to obtain the denoised signal. For VMD, the number of modes is set to $K = 10$, and the components are filtered based on their correlation coefficients with the original noisy signal. Only components with a high correlation are retained, while noise-dominant modes are removed.

The comparative results are summarized in Table 2.

Table 2. Performance comparison between the proposed FocuS-MN and representative baseline models.

Method	SDR (dB)	SDRi (dB)	SSNR (dB)
Noisy (Input)	−4.838	–	−3.699
EMD	−10.600	−5.762	−4.072
VMD	−7.852	−3.014	−3.897
DCCRN	2.379 ± 0.003	7.217 ± 0.003	2.478 ± 0.005
SEGAN	−0.359 ± 0.011	4.479 ± 0.011	0.282 ± 0.002
FullSubNet	2.023 ± 0.001	6.861 ± 0.001	2.180 ± 0.003
Ours (FocuS-MN, M = 6, N = 6)	**3.773** ± 0.002	**8.611** ± 0.002	**3.828** ± 0.001

As indicated in the table, traditional methods exhibit varying degrees of performance degradation in low-SNR underwater environments. Specifically, the SDR of EMD drops to −10.60 dB, yielding a negative gain (SDRi = −5.76 dB). Similarly, the SDRi of VMD is as low as −3.01 dB. This suggests that when processing highly non-stationary underwater acoustic signals, traditional methods not only fail to suppress noise effectively but also

introduce additional signal distortion. In contrast, the proposed FocuS-MN model achieves a significant improvement across all key metrics. Regarding the SSNR indicator, which reflects the quality of signal detail recovery, FocuS-MN improves the performance from −3.70 dB (noisy state) to 3.83 dB, representing an advantage of approximately 7.72 dB over the best-performing traditional method (VMD).

5.2.2. Comparison with Deep Learning Baselines

To comprehensively evaluate the performance of the proposed method, three representative deep learning models were selected as comparative baselines to cover diverse modeling paradigms. DCCRN adopts a complex domain time–frequency modeling strategy to achieve enhancement by jointly modeling magnitude and phase information, utilizing multi-layer LSTMs to capture long-range temporal dependencies within high-dimensional complex feature sequences—a strategy extensively validated in speech enhancement tasks. FullSubNet combines full-band and sub-band networks to incorporate global context at the sub-band level, effectively characterizing inter-band correlations. In contrast, SEGAN serves as an end-to-end time domain approach, utilizing a Generative Adversarial Network to directly learn the mapping from noisy to clean signals in the waveform domain, representing a modeling philosophy that bypasses explicit time–frequency transformations. These methods exhibit significant differences in structural design and modeling logic, providing a multi-faceted validation of the proposed FocuS-MN.

During the experimental phase, the three baseline models were retrained using the dataset described in Section 4.1. Specifically, DCCRN was optimized via the SI-SNR loss function, while the SEGAN generator was trained using a joint loss composed of Binary Cross-Entropy (BCE) adversarial loss to constrain distribution consistency and time domain L1 reconstruction loss to improve waveform fidelity. FullSubNet was trained using an L1 loss based on the magnitude spectrum to minimize the discrepancy between predicted and clean spectral representations. All baseline models utilized the Adam optimizer with a unified learning rate of 5×10^{-5} and were trained until the validation loss stabilized, after which the checkpoints with the highest validation performance were selected for testing.

To evaluate the practical deployability of the proposed model and baselines, we conduct a comprehensive profiling of their computational requirements. As summarized in Table 3, the total number of parameters and the floating point operations (FLOPs) are measured using a single-channel 3-second audio input at a sampling rate of 16 kHz, with a batch size of one to simulate real-time inference scenarios. The metrics were obtained using the fvcore library. It should be noted that the reported FLOPs represent a conservative estimate, as certain element-wise operations and specialized audio transforms (e.g., STFT/ISTFT and LSTM cells in DCCRN) may not be fully captured by the profiling tool due to their implementation-specific kernels. Nevertheless, the results provide a consistent relative benchmark for comparing the structural efficiency of the models.

The experimental results, as summarized in Table 2, indicate that the proposed FocuS-MN model achieved the optimal values across all three evaluation metrics. In terms of Signal-to-Distortion Ratio (SDR), FocuS-MN reached 3.7731 dB, representing a significant gain (SDRi) of 8.6114 dB over the noisy input. Furthermore, the model attained a Segmental SNR (SSNR) of 3.8271 dB, demonstrating a clear competitive lead over the second-best model. DCCRN emerged as the strongest baseline with an SDR of 2.3785 dB and an SSNR of 2.4775 dB; however, FocuS-MN outperformed it by 1.3946 dB in SDR and 1.3496 dB in SSNR. Meanwhile, FullSubNet and SEGAN achieved SDR values of 2.0225 dB and −3.0589 dB, respectively, further highlighting the robustness and superior noise suppression capabilities of the proposed architecture in complex underwater environments.

Table 3. Comparison of computational complexity.

Model	Params (M)	Trainable (M)	FLOPs (G)
DCCRN	2.69	2.69	11.34
FullSubNet	0.11	0.11	0.45
SEGAN (Generator)	0.18	0.18	1.37
SEGAN (Discriminator)	0.08	0.08	0.60
Ours (FocuS-MN, M = 6, N = 6)	0.34	0.34	18.73

5.2.3. Case Analysis

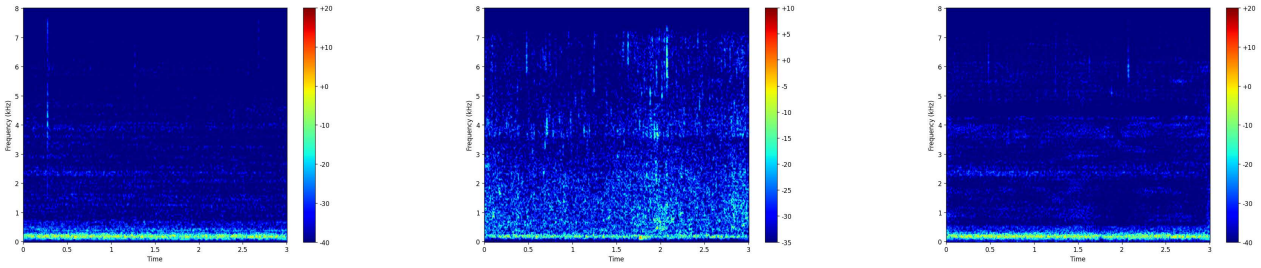
To further verify the practical performance of the FocuS-MN model in complex underwater environments, this section presents a single-sample analysis of a representative pilot boat-radiated noise signal selected from the test set. Pilot boats exhibit distinct line spectral features and harmonic structures in the low-to-medium frequency bands; therefore, processing this sample serves as an effective test of the model's ability to recover fine line spectral features while suppressing broadband noise.

Figure 6 illustrates the spectrograms of the sample at different stages:

Figure 6a, Clean Signal: Clearly displays the signature harmonic structures and fundamental frequency components of the pilot boat.

Figure 6b, Noisy Signal: Shows the low-frequency line spectral heavily masked by broadband background noise after the addition of marine ambient noise.

Figure 6c, FocuS-MN ($M = 6, N = 6$) Result: It can be observed that the model effectively eliminates broadband background noise while accurately restoring components below 500 Hz. Furthermore, for harmonic components above 1 kHz—specifically those around 2 kHz and 4 kHz—the results demonstrate the powerful reconstruction capability of the FSMN combined with the resampling module for long-range periodic features.

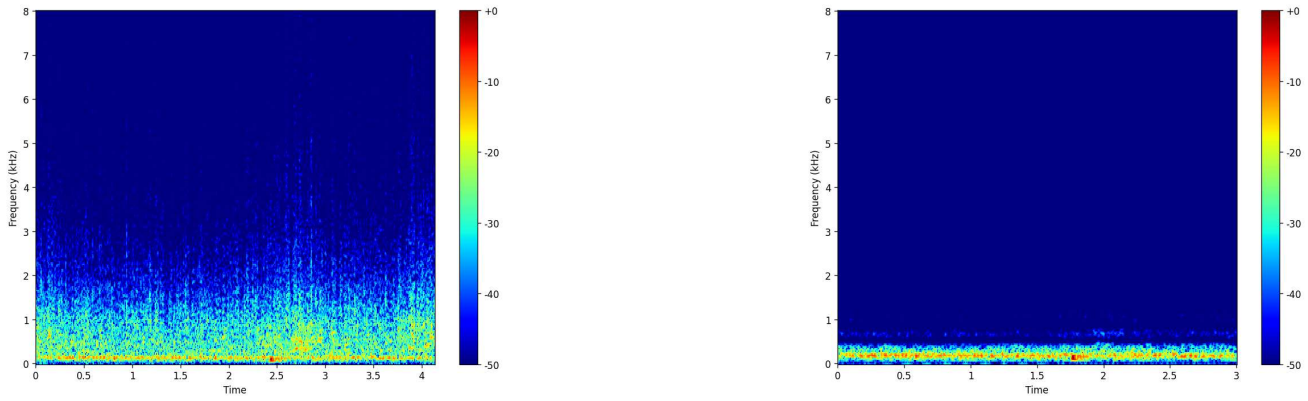


(**a**) Clean signal (**b**) Noisy signal (−5 dB) (**c**) Enhanced by FocuS-MN

Figure 6. Spectrogram visualization for comparison: (**a**) the ground truth clean signal; (**b**) the noisy signal at −5 dB SNR; (**c**) the enhanced signal reconstructed by the proposed FocuS-MN framework. The visualization demonstrates a significant reduction in broadband noise while preserving key harmonic components.

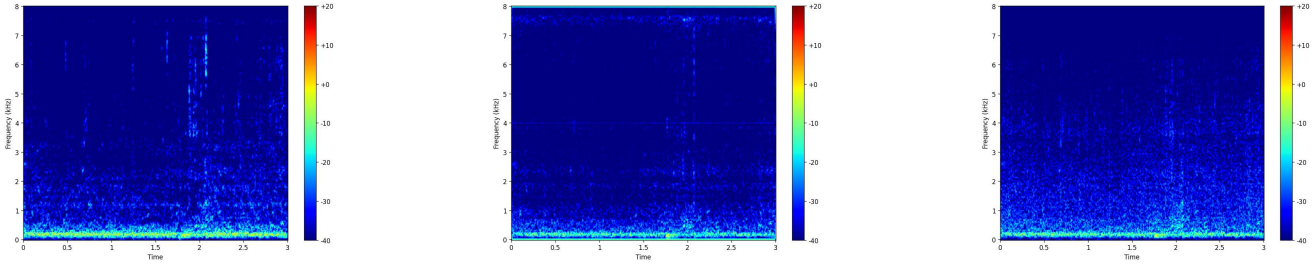
In addition, Figures 7 and 8 provide a cross-comparison with the results of traditional signal processing methods and mainstream deep learning approaches: traditional time–frequency analysis (VMD and EMD). As shown in Figure 7a,b, EMD fails to effectively suppress the non-stationary impulsive components in the signal. While VMD exhibits better band-pass characteristics, it tends to filter out key narrow-band harmonic components of the target signal as noise, resulting in significant feature loss. Deep learning baselines: SEGAN (Figure 8b) introduces low-frequency attenuation, damaging target energy below 500 Hz. DCCRN (Figure 8a) performs relatively well in line spectral recovery but fails to effectively remove the high-frequency noise occurring around the 2 s mark. FullSubNet (Figure 8c) fails to eliminate the broadband noise at 2 s and introduces interference into the low-frequency line spectral features.Compared to these methods, FocuS-MN not only

excels in Signal-to-Noise Ratio gain but also achieves the highest spectral fidelity, most closely resembling the original clean signal.



(**a**) EMD decomposition (**b**) VMD decomposition

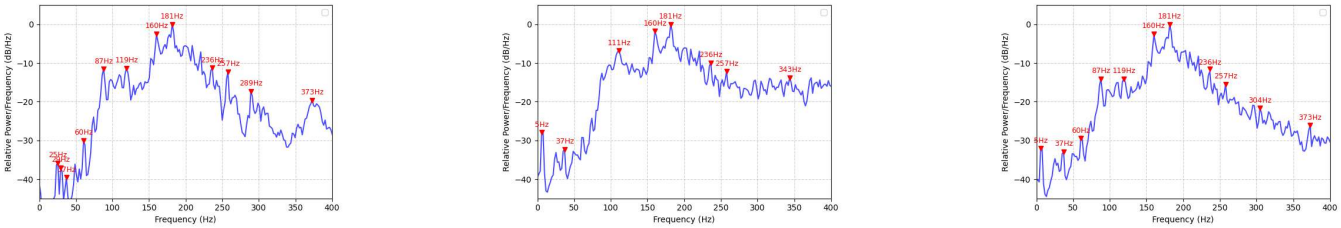
Figure 7. Comparison of traditional signal decomposition results for underwater acoustic signals: (**a**) Empirical Mode Decomposition (EMD) and (**b**) Variational Mode Decomposition (VMD). Both methods are employed as baselines to illustrate the limitations of classical non-stationary signal processing in high-noise environments.



(**a**) DCCRN (**b**) SEGAN (**c**) FullSubNet

Figure 8. Visual comparison of denoising results from different deep learning baselines: (**a**) DCCRN; (**b**) SEGAN; (**c**) FullSubNet. These spectrograms illustrate the varying degrees of noise suppression and spectral detail preservation across current state-of-the-art architectures.

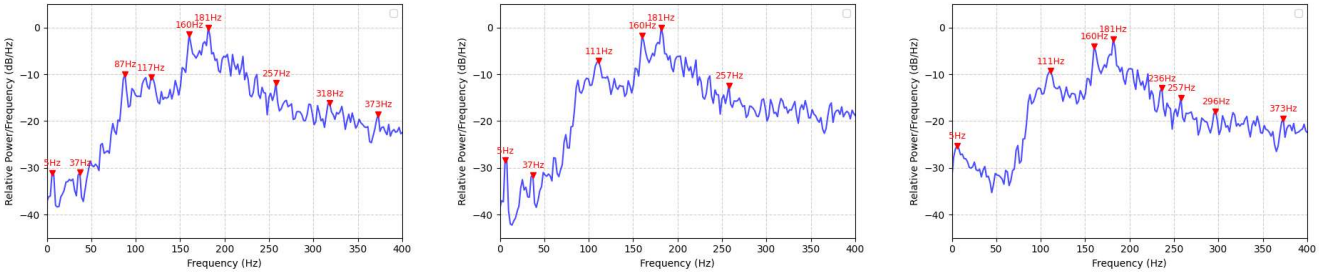
To evaluate the reconstruction fidelity of acoustic signatures, a peak-normalized PSD analysis was conducted (Figure 9). In the figure, the blue curves represent the PSD results, while the red labels indicate the frequency values of corresponding peaks (rounded to the nearest integer). The results indicate that at a low SNR of -5 dB, the dense line spectral within the 0–200 Hz band of the original signal are almost entirely masked by noise. In contrast, FocuS-MN successfully recovers these submerged spectral structures. Notably, at key frequencies such as 37 Hz, 60 Hz, 87 Hz, and 119 Hz, the reconstructed peaks align precisely with the clean reference. This demonstrates the model's robust capability to suppress broadband background noise while maintaining the high-fidelity preservation of low-frequency line spectral features.



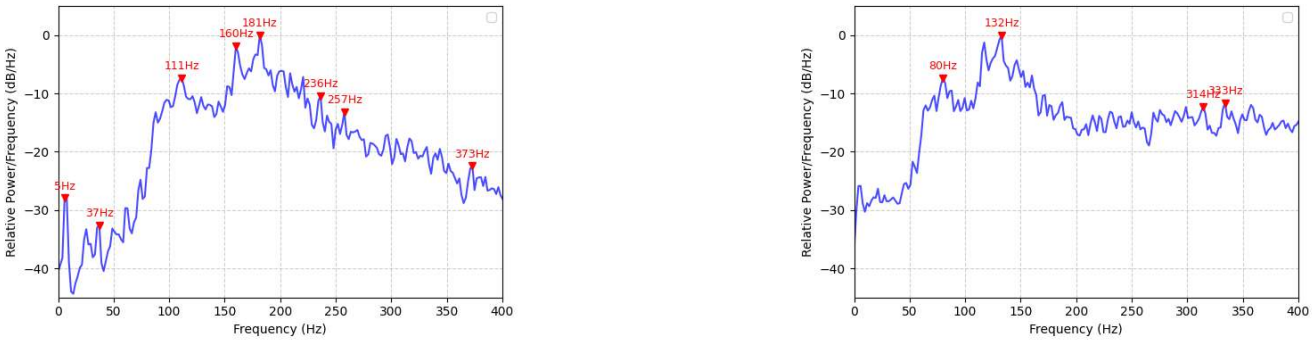
(**a**) Clean Signal (**b**) Noisy Signal (−5 dB) (**c**) Enhanced by FocuS-MN

Figure 9. PSD analysis of the proposed FocuS-MN under −5 dB SNR: (**a**) clean signal; (**b**) noisy input; (**c**) enhanced result. The comparison demonstrates the model's ability to recover submerged line spectral features and restore the original acoustic signature from heavy interference.

Furthermore, the PSD results of various baseline methods within the 0–200 Hz band—including both representative deep learning models and conventional signal processing techniques—are compared in Figure 10. The comparison reveals that conventional methods struggle to isolate spectral peaks from the ambient noise floor due to their limited robustness against non-stationary interference. While existing deep learning baselines achieve partial noise suppression, they often suffer from significant distortion or the complete loss of fine-grained line spectral features. In contrast, FocuS-MN demonstrates a superior decoupling performance, accurately reconstructing the original spectral distribution of the target vessel.



(**a**) Enhanced by DCCRN (**b**) Enhanced by FullSubNet (**c**) Enhanced by SEGAN



(**d**) Enhanced by VMD (**e**) Enhanced by EMD

Figure 10. PSD-based comparison of various baselines under low-SNR conditions: (**a**) DCCRN; (**b**) FullSubNet; (**c**) SEGAN; (**d**) VMD; (**e**) EMD. The results illustrate the performance variations in noise suppression and line spectral preservation between deep learning architectures and conventional decomposition methods.

As shown in Figure 11, we also compared the time domain waveforms of the noisy signal and the denoised signal. The thin light-red line represents the noisy signal, which contains a substantial amount of spikes and random perturbations. These components manifest as violent amplitude jumps in the time domain, masking the true periodic mechanical vibrations of the vessel. After processing with FocuS-MN (represented by the thick dark-blue line), these random transient noises were significantly suppressed, and the waveform became smooth and continuous. This demonstrates the model's excellent filtering capability against high-frequency uncorrelated noise.

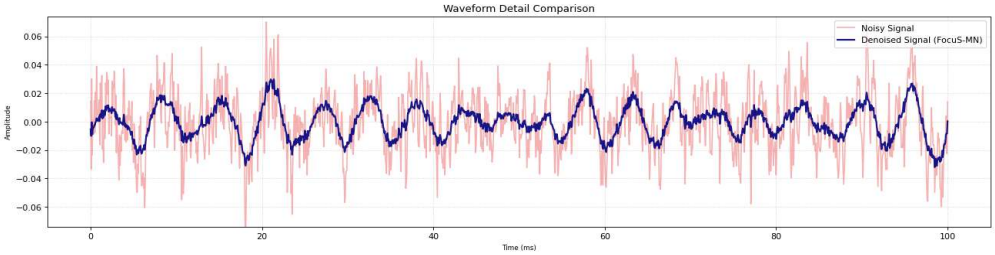


Figure 11. Detailed comparison of time domain waveforms. The plot illustrates a segment of the noisy signal (light red) overlaid with the denoised signal (dark blue) processed by FocuS-MN. The visualization highlights the model's effectiveness in filtering out high-frequency stochastic noise and transient impulses while accurately reconstructing the underlying periodic oscillations and structural envelope of the original vessel-radiated noise.

6. Discussion

6.1. Discussion of Ablation Study Results

The ablation results presented in Section 5.1 demonstrate that the performance of the FocuS-MN model is highly dependent on the deep synergy between the number of resampling layers (M) and the latent space modeling layers (N). The experimental data reveals that the model achieves an optimal balance in both SDR and SSNR metrics (3.77 dB and 3.82 dB, respectively) when both M and N are configured to six layers.

By analyzing the experimental results of the $\{M = 4, N = 4\}$, $\{M = 4, N = 6\}$, and $\{M = 4, N = 8\}$ groups, we observe a significant upward trend in performance metrics as the parameter N increases. This phenomenon indicates that the holistic stacking of resampling modules makes a prominent positive contribution to the model's temporal modeling performance. These stacking operations of downsampling and upsampling in the time–frequency domain are not merely means of computational optimization; rather, they continuously strengthen the model's ability to perceive and model the "time–frequency correspondence" within noisy audio signals.

In a further comparison of the $\{M = 4, N = 4\}$, $\{M = 6, N = 4\}$, and $\{M = 8, N = 4\}$ groups, a clear degradation in metrics is observed in the $\{M = 8, N = 4\}$ configuration. Our analysis suggests that this may be attributed to a "Buckets Effect" that occurs when the number of resampling layers significantly exceeds the stacking depth of the FSMN representation modules. Specifically, when the resampling feature resolution is insufficient, simply increasing the depth of the latent space may lead to a limited receptive field. Consequently, the model becomes unable to effectively capture global noise patterns, leading to a decline in overall performance.

Conversely, when comparing $\{M = 4, N = 4\}$ to $\{M = 6, N = 6\}$, we observe a more significant performance leap relative to other experimental groups (with a 0.21 dB increase in SDR). This suggests that symmetrical deepening simultaneously reinforces the multi-scale feature extraction capability and the modeling depth of long-range temporal dependencies. By maintaining a balanced architecture, the model ensures that the resolution

of processed features remains synchronized with the modeling capacity of the latent space, thereby maximizing the architectural stacking gain.

6.2. Potential for Scaling to High-Sampling-Rate Underwater Acoustic Signals

While the current experiments are primarily based on the ShipsEar dataset with a unified sampling rate of 16 kHz to verify fundamental performance, actual maritime environments contain ship-radiated noise and transient impulses with far richer high-frequency components. The architectural design of FocuS-MN specifically avoids compression along the frequency dimension within the encoder, a deliberate choice intended to preserve high-frequency details and capture sharp spectral transitions. This structural characteristic provides a robust foundation for the model to process underwater acoustic signals at significantly higher sampling rates, such as 44.1 kHz or 48 kHz.

Furthermore, the current downsampling and upsampling modules utilize fixed ratios (compressing T, F to $T/2, F/2$). To enhance processing flexibility for diverse audio sampling rates, these ratios can be adjusted to ensure optimal performance. Future research may introduce variable or adaptive resampling rates; for instance, by employing flexible downsampling ratios, FocuS-MN can significantly compress the sequence length processed by the FSMN blocks while maintaining the depth of feature representation. Simultaneously, adjusting the upsampling ratios ensures the integrity of spectral reconstruction for high-sampling-rate signals.

Consequently, future studies will test the model's generalization capabilities on broadband, high-sampling-rate data. This will allow us to evaluate whether the model can more flexibly focus on critical frequency bands of the target signal, thereby extending its utility to more complex and demanding underwater acoustic environments.

7. Conclusions

This paper proposes FocuS-MN, an end-to-end underwater target denoising framework specifically designed for the low Signal-to-Noise Ratio (SNR) conditions inherent in underwater acoustic environments. The model aims to "focus" on target ship signals amidst intense background interference by compressing time–frequency information through resampling and modeling latent space features via temporal sequences. This approach achieves effective noise suppression and signal reconstruction while maintaining structural fidelity. At the architectural level, FocuS-MN establishes a feature processing stream based on the joint input of magnitude and phase spectra. By employing a symmetrical resampling architecture, the model effectively reduces computational complexity while simultaneously enhancing its capability to represent latent space features. The integration of the FSMN (Feedforward Sequential Memory Network) allows for the focused learning of noise patterns within a lower-resolution space, followed by upsampling to restore structural details, resulting in a reliable high-fidelity signal reconstruction method.

The experimental results demonstrate that in extreme noise environments of -5 dB, FocuS-MN significantly outperforms traditional signal decomposition methods and several mainstream deep neural network models in generalization tests involving unseen vessel categories. Ablation studies confirm the synergistic effect between the resampling architecture and the FSMN modules. Notably, the symmetrically stacked $M = 6, N = 6$ configuration achieved an optimal overall performance, improving the Signal-to-Distortion Ratio (SDR) to approximately 3.77 dB and the Segmental SNR (SSNR) to approximately 3.82 dB.

Building upon these research findings and to further expand the potential of this method in the underwater acoustic field, future work will focus on the following directions:

High-Sampling-Rate Audio Scaling: We plan to extend the scope of our research to high-sampling-rate audio (44,100 Hz and above). By combining multi-scale resampling with FSMN modeling, we aim to achieve high-fidelity restoration of full-band signals, providing cleaner acoustic features for high-precision underwater target recognition.

Exploration of High Compression Ratios: Future experiments will further explore resampling factors of 4, 8, and 16. This will ensure effective noise modeling for high-sampling-rate audio even under extreme compression, further enhancing the model's representation capabilities and inference efficiency.

Robustness in Coupled Noise Scenarios: Considering the multi-source nature of marine noise, future research will incorporate fused datasets containing natural noise, short-term high-frequency impulses, and electrical noise from acquisition equipment. This will allow us to investigate the noise modeling performance of FocuS-MN in coupled noise scenarios, ultimately improving the framework's robustness and practical engineering value.

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Abbreviations

The following abbreviations are used in this manuscript:

FSMN	Feedforward Sequential Memory Network
SDR	Signal-to-Distortion Ratio
SSNR	Segmental Signal-to-Noise Ratio
PSD	Power Spectral Density
SNR	Signal-to-Noise Ratio
EMD	Empirical Mode Decomposition
VMD	Variational Mode Decomposition
DCCRN	Deep Complex Convolutional Recurrent Network
SEGAN	Speech Enhancement Generative Adversarial Network

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